

Room Connection Techniques in Virtual Reality for Walking in Impossible Spaces

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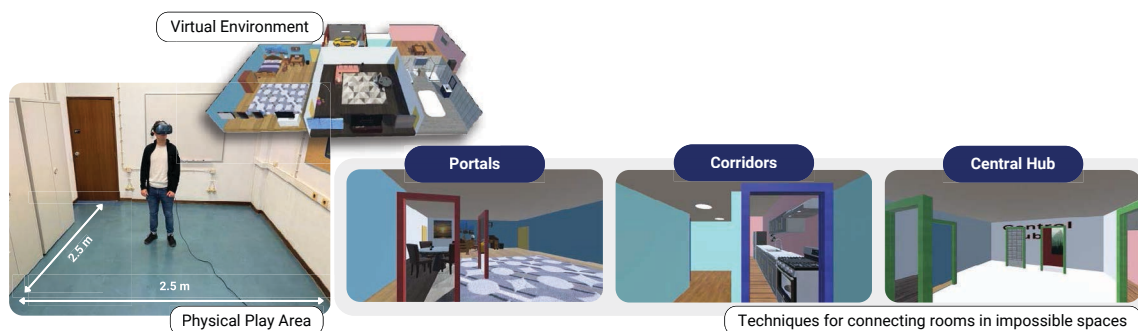
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Figure 1: Three room-connection techniques that enable VR walking-based navigation in a small physical spaces of about 2.5 x 2.5 meters play area to accommodate a five-room virtual apartment.

ABSTRACT

This work demonstrates three room-connection techniques – portals, corridors, and central hubs – that can be used in virtual environments to create “impossible spaces”. These spaces use overlapping areas to maximize available physical space, enabling walking even in constrained spaces. In this experiment, a user can navigate through a five-room apartment of about 106.25 m² by walking within a physical play area of about 6.25 m² (2.5 x 2.5 meters). The primary contribution of this work is demonstrating how these room-connection techniques can be applied to dynamically adapt virtual environments to fit small physical spaces, such as those commonly available to VR users at home. A video showcasing this demo can be found at the following link: https://youtu.be/a_4kcC0VOuY.

Index Terms: Virtual reality, impossible spaces, locomotion, navigation.

1 INTRODUCTION

Exploring a very large virtual environment (VE) naturally presents additional challenges when the available physical space is small. The average available physical area for Virtual Reality (VR) users is approximately 2.4 x 2.2 meters [1]. In more extreme cases, the play area may be as small as 1 x 1 meter, which represents the minimum requirement specified by Oculus¹. This limited physical space restricts the locomotion techniques available for exploring and navigating the VE. In such cases, the most commonly used locomotion techniques rely on handheld controllers for teleportation and joystick movements, compromising the natural sensation of walking and reducing the overall sense of immersion [5, 2]. Even

advanced techniques such as redirected walking [3], which rely on subtly adjusting users’ visual perception to keep them within physical boundaries, face challenges in very small areas [1]. These challenges can result in noticeable adjustments that may induce disorientation or motion sickness, highlighting the need for alternative solutions tailored to constrained physical environments.

We propose three techniques (Figure 1) – portals, corridors, and central hubs – to connect virtual spaces and maximize small physical spaces. These techniques allow users to navigate small physical spaces by overlapping virtual rooms, compressing large virtual worlds into the limited play areas available at home. This compression involves dividing the VE into smaller cells (e.g., decomposing an apartment into individual rooms), which are then connected using the proposed techniques. Additionally, adjustments can be made to the user’s stride if an individual room, even when decomposed, remains larger than the physical play area. By applying these connection techniques, a virtual layout is created where rooms overlap, resulting in a virtual world that defies the constraints of physical space, known as an “impossible space” [7, 6, 4].

2 ROOM CONNECTION TECHNIQUES

Building on the benefits of redirected walking, impossible spaces, and change blindness events, we propose our implementation of three techniques to connect virtual spaces: portals, corridors, and central hubs. All of these techniques maximize the use of physical space by creating overlapping areas in the VE. Each technique offers distinct advantages and addresses specific challenges in connecting rooms.

Portals provide a flexible and efficient way to connect rooms without requiring additional virtual elements, as they can be placed anywhere in the virtual space and are not confined to walls. Portals offer users a clear preview of the connected rooms, enhancing spatial awareness. However, portals require precise placement to avoid conflicts with other portals or objects in the VE.

Corridors preserve the natural flow of room connections and serve as transition zones, allowing smoother navigation between spaces. This makes them more familiar, as they appear as natural doorways within the walls of connected spaces. However, corridors

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¹ Setting up Guardian on Oculus: <https://www.meta.com/help/quest/articles/in-vr-experiences/oculus-features/boundary/> - Last accessed 28/01/2025

require additional virtual elements (i.e., adding walls to build the corridor). They also do not provide users with a preview of the next room, which may require more mental effort for users to create a mental map of the VE. Corridors may be impractical in very small play areas, such as 1 x 1 meter, as the space may not accommodate both the corridor and comfortable navigation.

Central hubs connect more than two rooms. This technique simplifies navigation, as a hub can link multiple rooms or even other hubs, making it useful for large, interconnected spaces (e.g., galleries). By acting as a centralized space, central hubs reduce the mental effort required by users, allowing them to easily return to the hub to choose their next room. However, this technique adds non-original elements (i.e., new rooms acting as hubs) to the VE that alter the original layout.

The room-connection techniques are employed to arrange a given VE to fit within the user's physical play area. The VE (e.g., a house) is first divided into smaller cells (e.g., rooms), which are then connected using one of the three proposed techniques. If the cells are still larger than the available physical space, we adjust the virtual user's stride within each cell to ensure the user can fully explore the entire area. For example, if a virtual room is twice as large as the physical space, the virtual user's stride is doubled.

3 EXPERIMENTAL SETUP

This experiment requires an unobstructed area of approximately 2.5×2.5 meters (Figure 1). The VE covers approximately 106.25 m², and users navigate the space by physically walking. A single controller is used to perform an object-finding task. The virtual world was developed using Unity Game Engine version 2020.3.3f1 and is designed to run on the HTC VIVE Cosmos headset. The experiment was conducted on a computer equipped with an NVIDIA GeForce RTX 4080 GPU with 16 GB of dedicated memory. However, any VR-ready computer should be capable of running the VE.

4 TASK

As a use case for this study, we selected virtual apartment tours as an example that can benefit the real estate industry by enabling remote visits to properties.

Participants searched for sixteen music vinyl LP record discs distributed across five rooms within the virtual apartment shown in Figure 1. The task aimed for simplicity and intuitiveness, allowing participants to focus on navigating the VE rather than managing additional cognitive challenges. Participants navigated physically through the apartment, guided by a visual cue on their virtual hand indicating the number of discs found (Figure 2). Upon locating a disc, participants interacted by touching the disc with their virtual hand to collect it. They then returned to the first room to place the disc into a music player, initiating playback throughout the virtual apartment. Participants continued until all sixteen discs were found, concluding the task.

5 CONCLUSION

The main contribution of this work is the introduction of three techniques as an approach to connect rooms by employing space overlapping to maximize the use of small physical spaces specifically. Supported by a previous study with 33 participants, the three techniques have been shown to provide comfortable navigation between rooms while highlighting their unique characteristics, making them applicable to various scenarios. Portals serve as a straightforward method for connecting rooms, offering a clear preview of the next room. This technique is suitable for scenarios where visual cues and immediate orientation are important, such as virtual museum tours or training simulations. Corridors maintain the original doors' positions and function as transitional areas, providing a smooth movement between rooms. This approach is ideal in cases where pre-

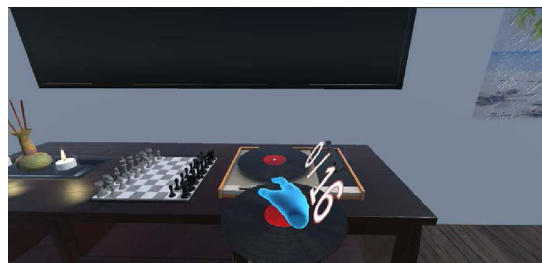


Figure 2: The user's task involved finding and placing sixteen vinyl LP record discs into a music player. Near the virtual hand, the user could see a counter indicating how many discs remained to be found.

serving the original entry and exit positions is crucial, such as virtual apartment tours. Central hubs allow multiple rooms to be connected through a central space, simplifying the navigation of more complex and large layouts. This technique is particularly effective for applications such as gallery visits or large-scale VEs.

Ultimately, we aim to enhance the walking experience in VR, specifically addressing the challenge of small physical spaces typically available in home settings. Our goal is to maximize the potential of VR, ensuring a more immersive and engaging experience despite spatial constraints.

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